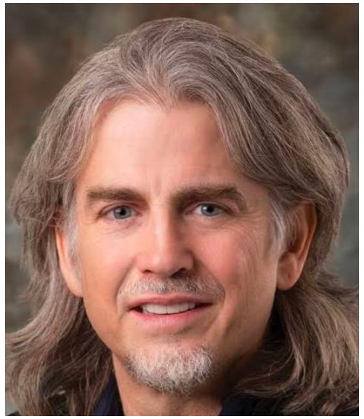
Introduction to Python

https://tinyurl.com/hbc-python

Harvard Chan Bioinformatics Core

Introductions!

Shannan Ho Sui
Director



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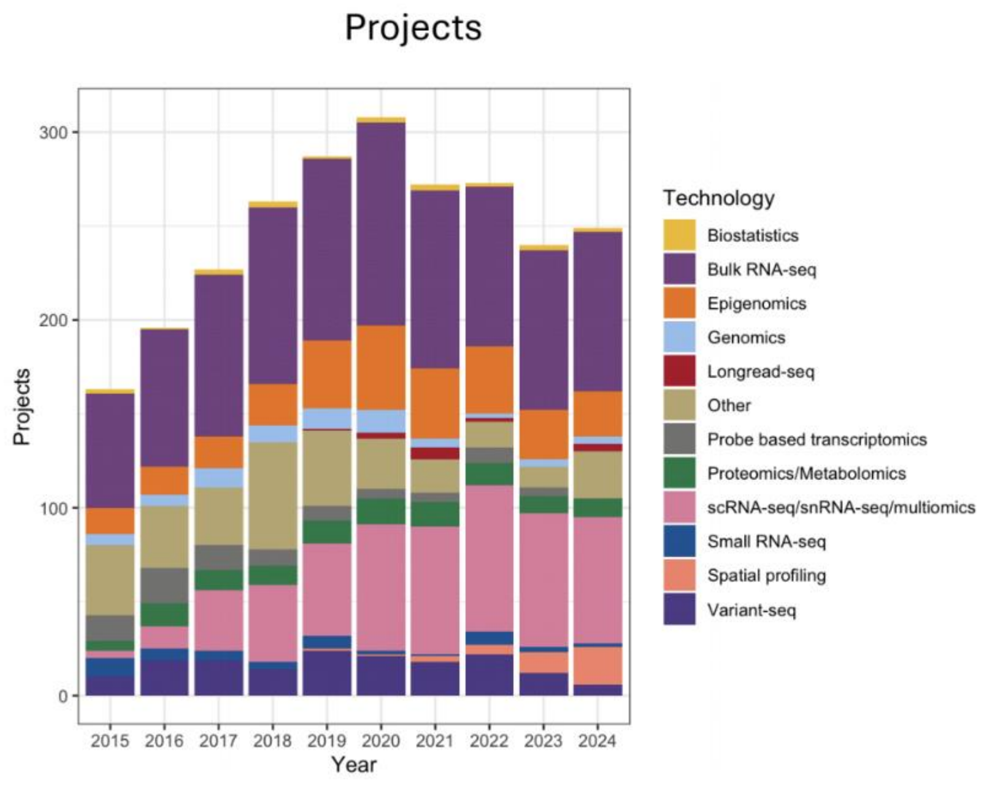
Consulting

- Transcriptomics: Bulk, single cell, small RNA
- Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- Variant discovery: WGS, resequencing, exome-seq and CNV
- Multiomics integration
- Spatial biology
- Experimental design and grant support



https://hsph.harvard.edu/research/bioinformatics/consulting/

Consulting

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HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH

NIEHS

HARVARD CATALYST

THE HARVARD CLINICAL AND TRANSLATIONAL SCIENCE CENTER

HARVARD MEDICAL SCHOOL

Training

- Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design
  - Basic Data Skills
    - Shell
    - R
    - Python
  - Advanced Topics: Analysis of high-throughput sequencing data
    - Chromatin Biology
    - Bulk RNA-seq
    - Differential Gene Expression
    - scRNA-seq
    - Variant Calling
  - Current Topics in Bioinformatics

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HARVARD UNIVERSITY CENTER FOR AIDS RESEARCH

DF/HCC DANA-FARBER / HARVARD CANCER CENTER

HARVARD MEDICAL SCHOOL

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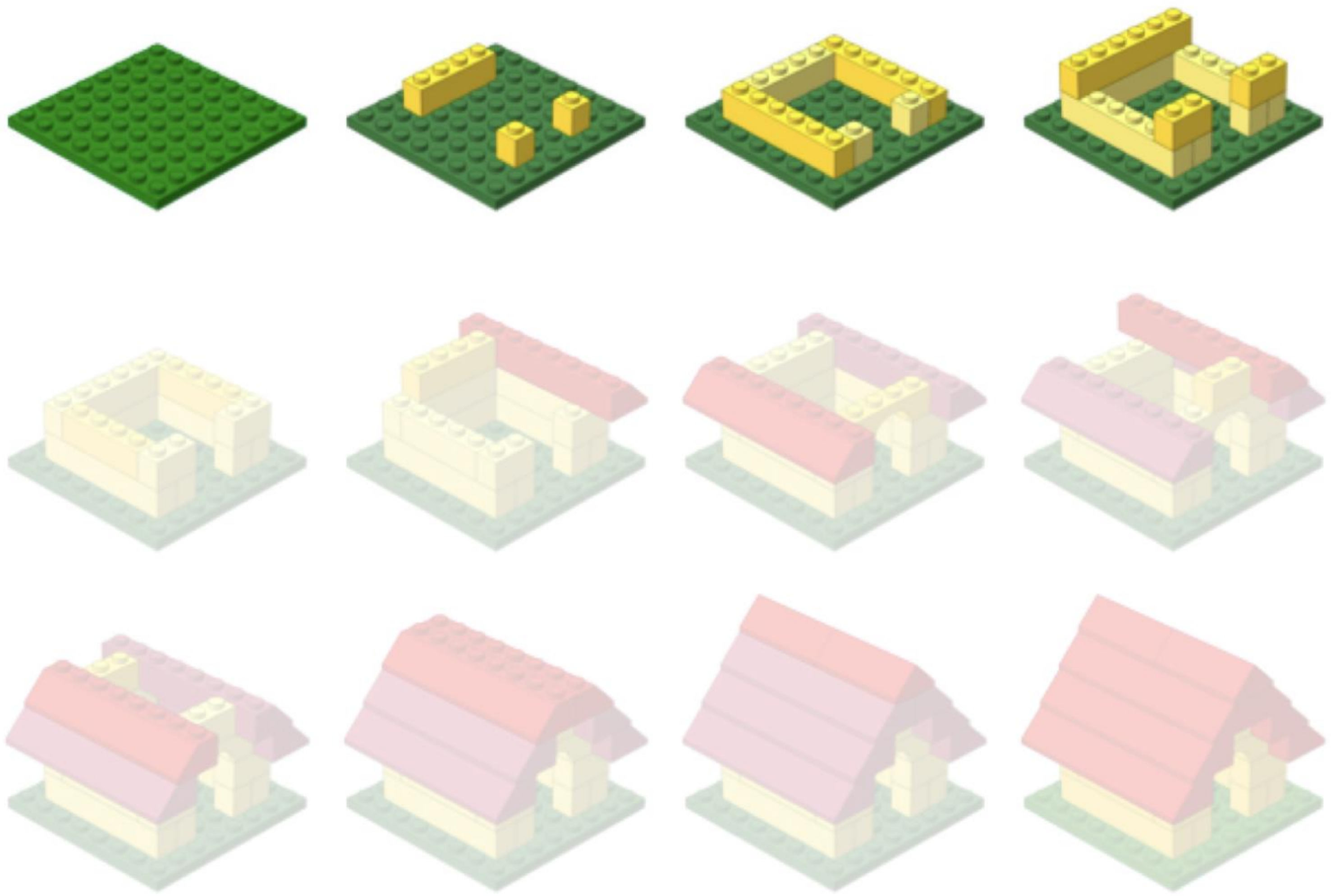
Workshop scope



Learning Python

Workshop Scope

- Setup and install Python and Jupyter Lab
- Write basic Python code and understand fundamental programming concepts
- Load and analyze data using Python libraries and functions
- Wrangle real-world datasets
- Create clear visualizations with Matplotlib and Seaborn
- Train a simple machine learning model using scikit-learn


Logistics

Course schedule

Introduction to Python Schedule HBC GitHub Contact us

Schedule - Introduction to Python

Code

Day 1

Time	Topic	Instructor
10:00 - 10:15	Workshop introduction	Will
10:15 - 10:45	Introduction to Python and Jupyter Lab	Will
10:45 - 10:50	Break	
10:50 - 11:25	Variables	Noor
11:55 - 12:00	Overview of self-learning materials and homework submission	Will

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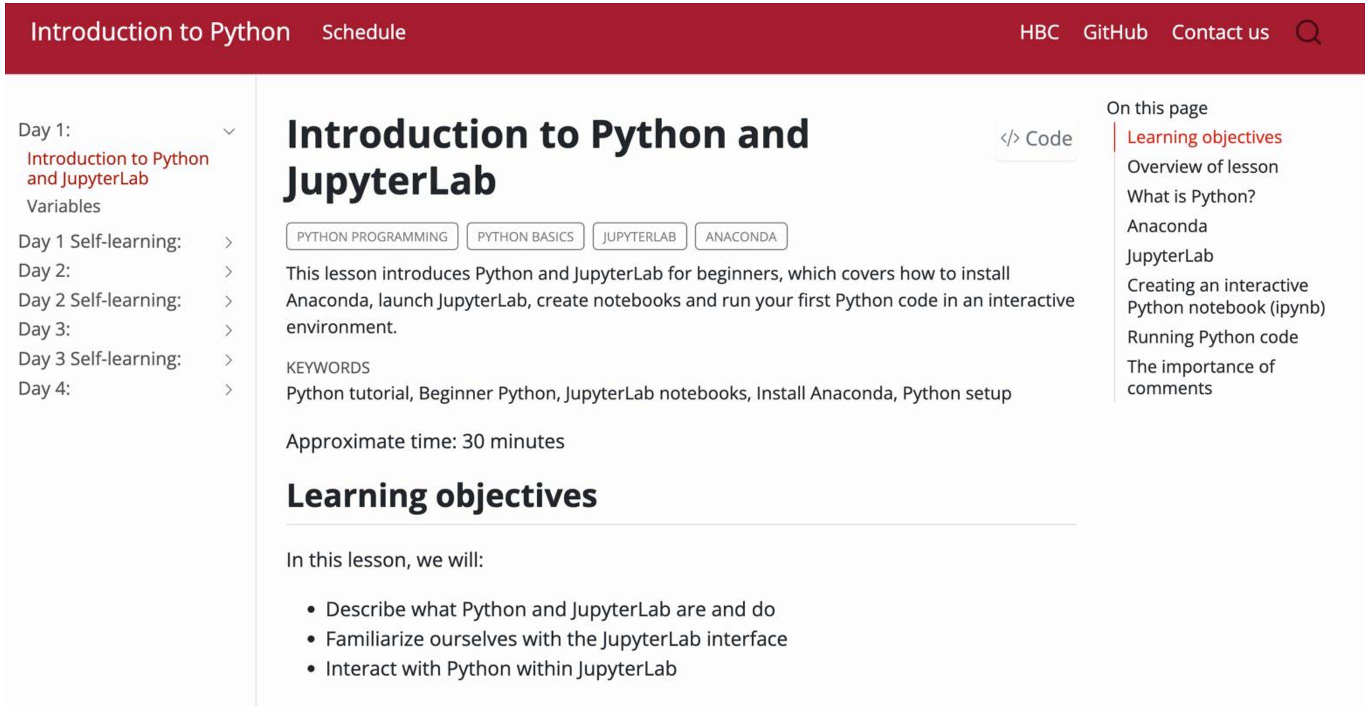
Course materials

❖ We continuously update our materials to reflect changes in the field/software



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Course participation

- Mandatory review of self-learning lessons and assignments
- Attendance required for all classes
- Your questions and active participation drive learning
- **We look forward to all your questions!**

Course participation

- At-home lessons and exercises after each session
- Cover material not previously discussed
- Provides us feedback to help pace the course appropriately
- 3-5 hours to complete
- Homework load is heavier near the end of the workshop

Using AI for Assignments

- Do
  - Try to resolve error messages with it
  - Test code written by AI on a dataset where you have expected results
  - Take the time to review the generated code line-by-line
- Don’t
  - Implement it in replacement to learning
  - Write code that you don’t understand
  - Assume the output from an AI process is correct

Odds & Ends

- Quit/minimize all applications that are not required for class
- Name tags
- Post-its
  - green - I am all set
  - red - I need time/help
- Phones on vibrate/silent
- Bathrooms

Contact Us

- *HBC training team:* hbctraining@hsph.harvard.edu
- *HBC consulting:* bioinformatics@hsph.harvard.edu